

Test Case Document

Feature: E-commerce Add to Cart, Checkout, Payment, Security, Usability, and Business Logic Validation

1. Document Information

Field	Details
Document Name	E-commerce Add to Cart and Checkout Test Case Document
Feature / Module	Product Detail Page, Cart, Checkout, Payment, Authentication, Order Creation
Prepared By	QA / Test Engineer
Test Type	Functional, Business Logic, Security, Usability, Regression, Analytics
Environment	Staging / QA
Status	Draft
Total Test Cases	55

2. Objective

The objective of this test case document is to validate the complete e-commerce purchase journey starting from product selection and add to cart, followed by cart calculation, coupon application, checkout, authentication, 2FA/security validation, payment, order creation, inventory update, analytics tracking, and post-order consistency.

The goal is to ensure that the user can complete a valid purchase successfully and that the system handles incorrect inputs, business rule violations, security risks, payment failures, usability issues, and cross-system data consistency correctly.

3. Scope

This document covers the following areas:

- Product variant selection
- Add to cart functionality
- Cart quantity update and removal
- Cart persistence after login

- Coupon and discount validation
 - Shipping and tax calculation
 - Grand total calculation
 - Checkout flow
 - Guest checkout and logged-in checkout
 - 2FA/security validation
 - Payment success, failure, timeout, and duplicate payment prevention
 - Inventory validation
 - Business rules such as free shipping, coupon limits, and taxability
 - Usability and accessibility checks
 - Analytics event validation
 - Order, payment, inventory, invoice, and email reconciliation
 - Cross-browser regression validation
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4. Out of Scope

The following areas are not covered in this document:

- Admin product creation
 - Warehouse fulfillment process
 - Refund processing after order completion
 - Return management workflow
 - Customer support workflow
 - Loyalty program or reward points, unless connected to checkout calculation
 - Real payment card processing in production
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5. Assumptions

The following assumptions are used while preparing these test cases:

- The product catalog contains at least one in-stock product.
 - The product supports variants such as size and color.
 - The cart supports quantity update, item removal, coupon application, shipping calculation, and tax calculation.
 - Payment gateway sandbox is available.
 - Test cards are available for successful, failed, and timeout payment scenarios.
 - Tax is calculated after discount unless business rules define otherwise.
 - Shipping fee depends on address, shipping method, product weight, or free-shipping rule.
 - User login, guest checkout, and 2FA behavior follow business configuration.
 - Backend recalculates price, tax, shipping, and discount; frontend values are not trusted.
 - Analytics tracking is enabled in the QA/staging environment.
 - Order confirmation email is enabled or can be verified through email logs.
 - Inventory is reduced only after successful order confirmation.
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6. Test Environment

Area	Details
Environment	Staging / QA
Browsers	Chrome, Safari, Firefox, Edge
Devices	Desktop, mobile web, tablet if supported
OS	Windows, macOS, iOS, Android
Payment Mode	Sandbox
Analytics Tools	GA4 / DebugView / BigQuery / Internal event logger
API Monitoring	Browser DevTools / Charles Proxy / Proxyman / Postman
Test Management Tool	Jira / TestRail / Zephyr / Xray / Google Sheets
Defect Tracking Tool	Jira / Linear / Azure DevOps / GitHub Issues

7. Test Data

Data Item	Value
Product Name	T-shirt
Product ID	TSHIRT-001
Variant	Black, Medium
Unit Price	\$100
Available Stock	5 units
Quantity	2
Valid Coupon	SAVE10
Coupon Rule	10% discount, minimum cart value \$100
Invalid Coupon	INVALID10
Expired Coupon	EXPIRED10
Shipping Fee	\$5
Tax/VAT	5% after discount
Payment Method	Sandbox card
Successful Test Card	Payment gateway successful sandbox card

Data Item	Value
Failed Test Card	Payment gateway failed sandbox card
2FA Code	Valid and invalid test OTPs

Expected calculation for quantity 2:

Calculation Item	Formula	Amount
Subtotal	$\$100 \times 2$	\$200
Discount	$\$200 \times 10\%$	\$20
Taxable Amount	$\$200 - \20	\$180
Tax	$\$180 \times 5\%$	\$9
Shipping	Fixed fee	\$5
Grand Total	$\$180 + \$9 + \$5$	\$194

8. Test Case Format

Each test case contains:

Field	Meaning
Test Case ID	Unique test case identifier
Category	Functional area
Test Case	Short description of what is being tested
Preconditions	Required setup before execution
Steps	Actions performed by tester
Expected Result	Correct system behavior
Priority	Critical, High, Medium, Low
Status	Pass, Fail, Blocked, Not Run
Defect ID	Linked bug ticket if failed

9. Test Cases

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-001	Cart	Add available product to cart	Product is in stock	Open product page, select size/color, click Add to Cart	Product is added with correct variant, price, quantity, and image	High	Not Run	
TC-002	Cart	Add product without selecting required variant	Product has required size/color options	Click Add to Cart without selecting required variant	Product is not added; required selection error is shown	High	Not Run	
TC-003	Cart	Add out-of-stock product	Product inventory is 0	Open product page and try Add to Cart	Add to Cart is disabled or out-of-stock message appears	High	Not Run	
TC-004	Cart Calculation	Quantity update calculation	Product added to cart	Change quantity from 1 to 3	Subtotal updates as unit price × quantity	High	Not Run	
TC-005	Inventory Rule	Quantity exceeds stock	Product stock is 5	Add product, change quantity to 6	System blocks update and shows maximum quantity message	High	Not Run	
TC-006	Cart	Remove product from cart	Product exists in cart	Click remove	Product is removed; subtotal, discount, tax, shipping, and total recalculate	Medium	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-007	Cart Persistence	Cart persists after login	Guest user has cart items	Add product, log in	Cart remains intact or merges correctly with saved cart	High	Not Run	
TC-008	Cart	Duplicate product add behavior	Same product and variant already in cart	Add same item again	Quantity increments instead of duplicate line item, unless business rule says otherwise	Medium	Not Run	
TC-009	Cart	Different variant behavior	Same product available in different size/color	Add two different variants	Cart shows separate line items for each variant	Medium	Not Run	
TC-010	Coupon	Valid coupon applies correctly	Valid coupon available	Add product, apply coupon	Discount is applied based on coupon rule	High	Not Run	
TC-011	Coupon	Invalid coupon rejected	Invalid coupon available	Apply invalid coupon	Coupon is rejected; total remains unchanged	High	Not Run	
TC-012	Coupon Business Rule	Coupon minimum order validation	Coupon requires \$100 minimum	Keep cart subtotal below \$100 and apply coupon	Coupon is not applied; minimum order error appears	High	Not Run	
TC-013	Tax	Tax/VAT calculation	Tax rule configured	Add taxable product and proceed to cart	Tax is calculated on correct taxable amount	High	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-014	Shipping	Shipping calculation	Address entered	Select delivery address	Shipping fee updates based on address, weight, delivery method, or free-shipping rule	High	Not Run	
TC-015	Total Calculation	Grand total accuracy	Product, coupon, tax, and shipping exist	Review cart total	Grand total equals subtotal - discount + tax + shipping	Critical	Not Run	
TC-016	Checkout	Checkout blocked for empty cart	Cart is empty	Click checkout	User cannot proceed; empty cart message appears	Medium	Not Run	
TC-017	Address	Address validation	Required address fields missing	Proceed to payment	System blocks checkout and highlights missing fields	High	Not Run	
TC-018	Payment	Payment success	Valid payment method available	Complete payment	Payment succeeds, order is created, cart is cleared	Critical	Not Run	
TC-019	Payment	Payment failure	Failed payment test card available	Submit payment	Payment fails; no confirmed order is created; cart remains intact	Critical	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-020	Payment Integrity	Double-click payment prevention	Payment page open	Click Pay button multiple times quickly	Only one payment request and one order are created	Critical	Not Run	
TC-021	Price Rule	Price change after cart addition	Admin changes product price after item is added	Add item, then refresh cart	System updates price with notice or preserves old price according to business rule	High	Not Run	
TC-022	Inventory Rule	Inventory change during checkout	Product stock becomes unavailable	Try to pay	Payment is blocked before charge, or order fails safely with clear message	Critical	Not Run	
TC-023	Coupon Persistence	Coupon after payment failure	Coupon applied and payment fails	Return to cart	Coupon remains or is removed according to business rule; totals stay consistent	Medium	Not Run	
TC-024	Payment Reliability	Payment timeout	Payment gateway timeout is simulated	Submit payment	Order status becomes pending/ failed safely; no duplicate charge occurs	Critical	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-025	Analytics	Analytics events accuracy	Analytics enabled	Add to cart, checkout, pay	Events fire once with correct product ID, price, quantity, discount, tax, shipping, and order ID	High	Not Run	
TC-026	Authentication	Guest checkout allowed	Guest checkout feature enabled	Add product, proceed to checkout without login	User can complete checkout as guest if business rule allows	High	Not Run	
TC-027	Authentication	Login required for checkout	Guest checkout disabled	Add product, proceed to checkout	User is redirected to login before payment	High	Not Run	
TC-028	Authentication	Cart preserved after forced login	Guest user has cart	Add item, proceed to checkout, log in	Cart items, coupon, and total remain unchanged after login	High	Not Run	
TC-029	Security / 2FA	2FA triggered for high-risk login	User has 2FA enabled	Log in during checkout from new device/ browser	2FA challenge appears before checkout continuation	Critical	Not Run	
TC-030	Security / 2FA	Correct 2FA code allows checkout continuation	2FA challenge visible	Enter valid 2FA code	User is authenticated and returned to checkout with cart preserved	Critical	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-031	Security / 2FA	Incorrect 2FA code rejected	2FA challenge visible	Enter invalid 2FA code	Access is denied; clear error appears; checkout does not proceed	Critical	Not Run	
TC-032	Security / 2FA	2FA retry limit enforced	2FA enabled	Enter wrong code repeatedly	Account/session is temporarily blocked or rate-limited according to security policy	Critical	Not Run	
TC-033	Security / Session	Checkout session expiry	User inactive during checkout	Wait until session timeout, then click Pay	User is asked to re-authenticate; payment is not submitted with expired session	Critical	Not Run	
TC-034	Security / Authorization	User cannot access another user's cart/order	Two user accounts available	Attempt to open another user's cart/order URL	Access is denied; no cart/order data is exposed	Critical	Not Run	
TC-035	Security / Price Tampering	Client-side price manipulation blocked	Browser dev tools/proxy available	Modify item price in frontend/API request and submit checkout	Backend recalculates price; manipulated price is rejected	Critical	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-036	Security / Quantity Tampering	Client-side quantity manipulation blocked	Stock is 5	Modify API request quantity to 999	Backend rejects quantity above stock/business limit	Critical	Not Run	
TC-037	Security / Coupon Abuse	Coupon cannot be reused beyond allowed limit	Coupon has one-use limit	Apply same coupon after previous successful order	Coupon is rejected if usage limit is reached	High	Not Run	
TC-038	Security / Payment	Payment callback validation	Payment gateway sandbox available	Simulate fake/invalid payment callback	Order is not confirmed unless callback/signature is valid	Critical	Not Run	
TC-039	Security / Sensitive Data	Payment details not stored or exposed	Payment completed	Inspect order page, logs, and API response	Full card number/CVV is never stored or exposed; only safe token/last four digits are shown if allowed	Critical	Not Run	
TC-040	Security / CSRF	Cart/payment actions protected from CSRF	User logged in	Attempt cross-site cart/payment request	Unauthorized request is rejected by CSRF/session protection	High	Not Run	
TC-041	Business Logic	Free shipping threshold	Free shipping rule configured at \$150	Test cart below and above \$150	Shipping becomes free only when eligible amount reaches threshold	High	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-042	Business Logic	Coupon and free shipping interaction	Coupon applied near free-shipping threshold	Apply coupon that reduces payable amount	Free shipping eligibility is calculated according to defined business rule	High	Not Run	
TC-043	Business Logic	Product category coupon restriction	Coupon valid only for selected category	Apply coupon to excluded category product	Coupon is rejected with correct reason	High	Not Run	
TC-044	Business Logic	Maximum order quantity per user	Product has max order limit	Try to buy more than allowed quantity	System blocks purchase above allowed quantity	High	Not Run	
TC-045	Business Logic	Mixed taxable and non-taxable items	Cart contains taxable and non-taxable products	Add both product types	Tax applies only to taxable items	High	Not Run	
TC-046	Business Logic	Returnable vs non-returnable product flag	Product is non-returnable	Complete checkout	Order confirmation and product page show correct non-returnable policy	Medium	Not Run	
TC-047	Usability	Add to cart feedback is clear	Product page loaded	Click Add to Cart	User sees clear confirmation, cart count updates, and next action is obvious	Medium	Not Run	
TC-048	Usability	Cart count updates correctly	Cart has 0 items	Add item, update quantity, remove item	Header cart count updates accurately at each step	Medium	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-049	Usability	Loading state during checkout	Checkout request takes time	Click Continue or Pay	Button shows loading state and prevents duplicate submission	High	Not Run	
TC-050	Usability	Error messages are understandable	Invalid coupon/ address/ payment used	Trigger errors	User sees specific, human-readable messages, not raw system errors	Medium	Not Run	
TC-051	Usability	Mobile responsive checkout	Mobile device or viewport available	Complete add to cart and checkout flow	Layout is usable; buttons, fields, totals, and payment CTA are visible and tappable	High	Not Run	
TC-052	Accessibility	Keyboard navigation works	Desktop browser available	Navigate cart and checkout using keyboard only	User can select variants, update cart, apply coupon, and proceed using keyboard	Medium	Not Run	
TC-053	Accessibility	Screen reader labels for checkout fields	Screen reader or accessibility inspector available	Inspect cart, address, coupon, and payment fields	Important fields and buttons have meaningful labels	Medium	Not Run	

ID	Category	Test Case	Preconditions	Steps	Expected Result	Priority	Status	Defect ID
TC-054	Data Consistency	Order, inventory, payment, and invoice reconciliation	Payment succeeds	Compare order DB, payment gateway, inventory, invoice, and email	All systems show same order ID, amount, currency, status, and item quantity	Critical	Not Run	
TC-055	Regression / Cross-Browser	End-to-end checkout across browsers	Chrome, Safari, Firefox/Edge available	Complete purchase flow on each browser	Add to cart, calculation, payment, and order confirmation work consistently	High	Not Run	

10. Detailed Critical Test Case

Test Case ID: TC-020

Field	Details
Title	Prevent duplicate order creation when user clicks Pay multiple times
Category	Payment Integrity / Race Condition
Priority	Critical
Severity	Critical

Preconditions

- User is logged in.
- Cart contains one product.
- Product is in stock.
- Payment gateway sandbox is enabled.
- User is on the payment page.

Test Data

Item	Value
Product Price	\$100
Quantity	2

Item	Value
Coupon Discount	10%
Shipping Fee	\$5
Tax/VAT	5%
Expected Grand Total	\$194

Steps

1. Add product `TSHIRT-001` to cart.
2. Set quantity to 2.
3. Apply coupon `SAVE10`.
4. Proceed to checkout.
5. Add valid shipping address.
6. Select card payment.
7. Rapidly click the Pay button 3–5 times.
8. Wait for payment response.
9. Check user order history.
10. Check payment gateway logs.
11. Check database order records.
12. Check inventory count.
13. Check confirmation email.

Expected Result

- Only one payment transaction should be created.
- Only one order should be created.
- Inventory should reduce once.
- User should receive one confirmation email.
- Cart should be cleared once.
- Pay button should be disabled after the first click.
- Backend should enforce idempotency so duplicate frontend requests cannot create duplicate orders.

Failure Conditions

The test fails if:

- More than one payment request is created.
- More than one order is created.
- Inventory is reduced more than once.
- User receives multiple confirmation emails.
- Cart/order state becomes inconsistent.

11. Analytics Validation Checklist

For this scenario, the following analytics events should be validated:

Event Name	Trigger	Key Parameters
<code>product_viewed</code>	User opens product page	product_id, product_name, price, category
<code>add_to_cart</code>	User adds item to cart	product_id, variant_id, quantity, unit_price, cart_id
<code>cart_updated</code>	User changes quantity/removes item	cart_id, product_id, old_quantity, new_quantity
<code>coupon_applied</code>	User applies valid coupon	coupon_code, discount_amount, cart_value
<code>coupon_failed</code>	Invalid coupon attempt	coupon_code, failure_reason
<code>checkout_started</code>	User starts checkout	cart_id, subtotal, discount, tax, shipping, grand_total
<code>payment_started</code>	User clicks Pay	order_id/checkout_id, amount, currency, payment_method
<code>payment_failed</code>	Payment fails	checkout_id, failure_reason
<code>purchase_completed</code>	Payment succeeds and order is confirmed	order_id, amount, currency, product_id, quantity, discount, tax, shipping
<code>order_confirmation_viewed</code>	Order success page appears	order_id, amount, status

Validation rules:

- Events should fire once per valid action.
- Failed payment should not trigger `purchase_completed`.
- `purchase_completed` should fire only after confirmed payment and order creation.
- Product ID, quantity, discount, tax, shipping, and total should match backend order data.
- No sensitive payment data such as full card number or CVV should be sent in analytics.

12. Defect Reporting Format

If any test case fails, report the bug using the following format:

Field	Details
Title	Clear one-line summary of the issue
Environment	Browser, OS, device, app/web version, environment
Preconditions	Required setup
Steps to Reproduce	Exact steps
Actual Result	What happened
Expected Result	What should have happened
Severity	Impact level
Priority	Urgency
Attachments	Screenshots, screen recording, logs, API traces
Failed Test Case ID	Example: TC-020
Defect ID	Jira/Linear/Azure DevOps ticket ID

13. Example Bug Report

Title

Multiple orders created when Pay button is clicked rapidly

Severity

Critical

Priority

P0

Environment

Field	Details
Browser	Chrome 121
OS	Windows 11
Environment	Staging
User Type	Logged-in customer
Payment Method	Sandbox card payment

Preconditions

- Product is in stock.
- User is logged in.
- Cart contains product `TSHIRT-001`.
- Payment gateway sandbox is enabled.

Steps to Reproduce

1. Add product worth \$100 to cart.
2. Set quantity to 2.
3. Apply 10% coupon.
4. Proceed to checkout.
5. Select card payment.
6. Rapidly click Pay multiple times.
7. Check order history and payment gateway logs.

Actual Result

- Two orders are created for the same cart.
- Inventory is reduced twice.
- Two payment requests are sent to the payment gateway.

Expected Result

Only one order and one payment request should be created for the checkout session.

Impact

This can cause duplicate charges, incorrect inventory reduction, refund requests, customer complaints, and financial reconciliation issues.

Likely Root Cause

Frontend does not disable the Pay button after the first click. Backend may not be enforcing idempotency for payment/order creation.

Recommended Fix

- Disable the Pay button immediately after the first click.
- Add backend idempotency key using cart ID, user ID, and checkout session ID.
- Reject duplicate payment/order requests for the same active checkout session.
- Ensure payment and order creation are handled atomically.

14. Execution Summary

This section should be filled after test execution.

Metric	Count
Total Test Cases	55
Passed	
Failed	
Blocked	
Not Run	
Defects Raised	
Critical Defects	
High Defects	
Medium Defects	
Low Defects	

15. Sign-off

Role	Name	Status	Date	Comments
QA Engineer				
QA Lead				
Product Manager				
Engineering Lead				

16. High-Risk Areas

The following areas should receive the most attention during release testing:

- Duplicate payment or duplicate order creation
- Incorrect final payable amount
- Client-side price or quantity tampering
- Inventory becoming negative or oversold
- Coupon abuse
- Order confirmed without valid payment
- Payment completed but order not created
- Order created but payment failed
- Cart lost after login or 2FA
- Incorrect analytics causing false business reporting

- Sensitive payment or customer data leakage
 - Poor checkout usability on mobile devices
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17. Final Notes

This test case document should be updated whenever business rules change. For example, changes in coupon policy, free shipping threshold, tax calculation method, guest checkout availability, 2FA rules, or payment gateway behavior should be reflected in the assumptions, test data, and affected test cases.